OSF magnetometry

One NMC pouch cell; five voltage anchors

41 scans/anchor; six magnetic channels

Separate sources not paired

PulseBat (retired 35-Ah LFP)

SOH and SOC conditioning variables

21 pulse-voltage features

OSF morphology bank

Five anchor means / standard deviations

Interpolated and sampled base morphology

Grounded-request inputs

v = mean of 21 pulse-voltage features; clipped to [3.06, 3.34] V; SOH and SOC from PulseBat

GROUNDED: 3.06–3.34 V

bridge v1.3 with synthetic SOH conditioning

- QRec embedding (171-D) → LDA projection (4-D)
- Signed synthetic SOH displacement in LDA space
- Cone-restricted 8-NN softmin retrieval of OSF scans
- SOH-dependent morphology blending with OSF base
- SOH-scaled amplitude jitter + spectral broadening
- Sensor noise + SOC-dependent low-frequency noise

REGIME-B

v = 2.54 / 2.81 / 3.00 V

Same bridge v1.3 code is called.
Internal SOH = 1.0:
- zero SOH displacement
- zero SOH-dependent blend
QRec/decoder still evaluated.
OSF base + sensor noise + PulseBat SOC-dependent noise.
Stored SOH is undefined (NaN).

Synthetic anomalies

Perturb 600 clean grounded parents

4 subtypes × 150; no stacking

Grounded clean
5,600 samples

Synthetic anomalies
600 samples

Regime-B
560 samples

MagBridge-Battery v1.0 | 6,760 synthetic samples
Six length-100 channels per sample + provenance metadata